
Prompt Engineering Challenge – Week 1 Assignment

1. Summarising Notes

Prompt

I am a B.Tech student preparing for semester exams. Summarize the following notes into easy-to-revise bullet points. Highlight important definitions, key concepts, and exam-focused points. Keep the summary clear, concise, and suitable for last-minute revision.

AI Response

Artificial Intelligence Revision Notes

- Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence.
- Machine Learning (ML) is a subset of AI that learns from data.
- Generative AI creates content such as text, images, and code.
- Agentic AI can make decisions and perform actions toward specific goals.
- AI applications include healthcare, education, finance, transportation, and cyber security.
- AI systems improve performance through data analysis and pattern recognition.

Why the Prompt Worked

The prompt clearly defined the audience (B.Tech student), purpose (exam preparation), and desired format (bullet points), resulting in a concise and useful summary.

2. Generating a Study Plan

Prompt

I am a B.Tech Computer Science student who wants to improve academic performance and prepare for placements. Create a 4-week study plan with 3 hours of study daily. Include programming, aptitude, communication skills, revision, and mock tests.

AI Response

Week 1

- Learn Python basics
- Practice aptitude arithmetic
- Read English articles for communication skills

Week 2

- Solve coding problems
- Practice reasoning questions
- Revise previous topics

Week 3

- Learn data structures
- Take aptitude mock tests
- Practice interview questions

Week 4

- Full revision
- Coding practice
- Placement preparation tests

Why the Prompt Worked

The prompt specified the student's background, goals, and available study time, helping the AI create a realistic and structured study plan.

3. Creating Interview Questions

Prompt

Act as a placement interviewer. Generate 10 interview questions and answers for a B.Tech Computer Science student applying for a software development internship. Include technical and HR questions.

AI Response

1. What is Python?

Python is a high-level programming language known for simplicity and readability.

2. What is Object-Oriented Programming?

A programming approach based on classes and objects.

3. Difference between List and Tuple?

Lists are mutable; tuples are immutable.

4. What is a Database?

A structured collection of data.

5. What is Git?

A version control system used to track code changes.

6. What are APIs?

Interfaces that allow applications to communicate.

7. Why should we hire you?

I am eager to learn, adaptable, and have strong problem-solving skills.

8. What is debugging?

Finding and fixing errors in code.

9. What is Machine Learning?

A branch of AI where systems learn from data.

10. What are your strengths?

Quick learner, teamwork, and analytical thinking.

Why the Prompt Worked

The prompt specified the job role, education level, and type of questions needed, resulting in relevant interview preparation content.

4. Writing a Resume Summary

Prompt

Write a professional resume summary for a B.Tech Computer Science student interested in Artificial Intelligence, Machine Learning, Software Development, and internships. Highlight technical skills, academic learning, teamwork, and career goals.

AI Response

Motivated B.Tech Computer Science student with a strong interest in Artificial Intelligence, Machine Learning, and Software Development. Skilled in Python, problem-solving, and collaborative teamwork. Passionate about learning emerging technologies and applying technical knowledge through projects and internships. Seeking opportunities to contribute to innovative solutions while continuously improving technical and professional skills.

Why the Prompt Worked

The prompt clearly described the student's profile, interests, and goals, enabling the AI to generate a targeted and professional resume summary.

5. Creating a Lesson Plan

Prompt

Create a 60-minute lesson plan for teaching Python loops to first-year B.Tech students. Include learning objectives, examples, classroom activities, practice exercises, and assessment questions.

AI Response

Learning Objectives

- Understand the purpose of loops
- Learn `for` and `while` loops
- Write simple loop-based programs

Activities

- Introduction to loops (10 minutes)
- Demonstration with examples (15 minutes)
- Guided coding practice (15 minutes)
- Individual exercises (15 minutes)
- Q&A session (5 minutes)

Practice Exercises

1. Print numbers from 1 to 10.
2. Print even numbers between 1 and 20.
3. Calculate the sum of numbers from 1 to 100.

Assessment Questions

- What is a loop?
- Difference between `for` and `while` loops?
- When should loops be used?

Why the Prompt Worked

The prompt specified the audience, topic, duration, and required sections, allowing the AI to generate a complete and organized lesson plan.